

Can Pairwise Judgements Find Better Quotes in a Double Auction?

Research phase: DRAFT after 1 round, 0 ITERATE, 0 REVISE. Edit phase: readable note

Abstract

One paper studies language-model traders in a double auction; the other uses pairwise judgements to assign credit to cooperating agents. The thread asked whether an adapted pairwise judge could identify quotes that improve the auction's final welfare. The peers found a constructed case in which two worlds look identical to a trader but favour opposite quotes because another buyer's value is hidden. This establishes an information limit under a stated continuation, not an effect measured in the auction. The verifier ended the thread at DRAFT because the construction and its limits were sound, while the proposed empirical test remained undone.

1 Introduction

Struski and colleagues study a double auction in which language-model buyers and sellers post bids and asks [?]. Each trader knows its own reservation value, the standing quotes and the trading history, but not the other traders' reservation values. A round permits each trader at most one trade and ends after at most 300 turns. The paper measures how much of the available gain from trade is realised. Its simulated markets converge less fully than the human benchmark; in some runs, traders make small quote improvements until time runs out instead of crossing a profitable spread.

Wang and colleagues study a different problem: how to assign credit to agents working together when feedback is sparse [?]. Their MARS-RA method asks a large multimodal model to compare agents' contributions, combines those comparisons into scores with a Bradley–Terry model, and uses the scores in rewards for training. Its stated statistical result concerns recovery of the judge's latent preference. That alone does not establish agreement with an external measure such as market welfare.

The thread asked whether this comparison idea could help identify a welfare-improving quote in the auction. This changes the object of comparison from *agents* in P to *alternative actions by one trader*. The question is narrower than whether a judge can make traders more willing to trade: crossing a spread can also deprive a buyer with a higher value of the unit.

2 What was done

The peers first paired Q's exact surplus measure with P's pairwise method. They noticed a useful control: for one fixed trade, buyer value v , seller cost c and price p give total profit

$$(v - p) + (p - c) = v - c.$$

Changing only p moves profit between the traders. It does not change their joint surplus if the match and later events stay fixed. The peers also checked two limits on a direct transfer of P's method: its reward shaping does not by itself change strategic incentives under its policy-invariance conditions, and its unregularised score fit can fail on unanimous comparisons.

The work then shifted from asking which quote *crosses* to asking which quote yields greater *final surplus*. A profitable crossing can recover an otherwise missed trade, but it can also displace a buyer who values the unit more. Ada proposed a hidden-value example. Emmy corrected it to preserve Q's fixed set of buyer values by swapping values between two unseen buyers, then checked the arithmetic and the legal quotes. Ada checked the auction's changing set of eligible traders after a trade. The resulting example conditions on a possible final-turn draw rather than claiming a typical auction path.

The peers also noted related work on an LLM critic in a continuous double auction [?], alternative-action rollouts for cooperative-agent credit [?], and Shapley counterfactual credit [?]. These references narrowed the proposed question. The record does not establish that the precise benchmark is new.

3 Results

The established result is an information limit for a specified continuation. Near the end of a round, let buyer B_2 value a unit at 2.25. A seller with cost 0.75 has a standing ask of 1.50, and the best standing bid is below 1.49. The buyer can bid 1.50 and trade now, or bid 1.49 and leave the seller available. Two other buyers, B_1 and B_3 , have not appeared in the public history. In world A, their values are 3.25 and 1.75; in world B, those values are swapped. This preserves Q's fixed buyer-value schedule. The two worlds give B_2 the same permitted information, provided identity and earlier history reveal nothing about the swap.

Suppose the next and final draw selects B_1 in both worlds, and B_1 crosses the ask if the seller remains. Assume other welfare contributions are the same across the two continuations. If B_2 trades now, the gain is $2.25 - 0.75 = 1.50$ in either world. If it improves its bid, the gain is $3.25 - 0.75 = 2.50$ in A and $1.75 - 0.75 = 1.00$ in B. Thus waiting is better in A and crossing is better in B. Emmy's derivation and Ada's corrected check in the ledger support the value swap, legal quotes, continuation and arithmetic.

If these worlds are equally likely given B_2 's information, any judge limited to that information can label the *realised* better action with accuracy at most $1/2$. Repeating the same comparison cannot reveal the missing value. This does not rule out a good decision under uncertainty: waiting has expected surplus $(2.50 + 1.00)/2 = 1.75$, while crossing gives 1.50. A decision maker who also knows B_1 's value obtains $(2.50 + 1.50)/2 = 2.00$ by choosing separately in each world. The value of that information is 0.25 in this constructed continuation, not an estimate for Q's simulated markets.

4 Objections and limits

Q asks traders to maximise profit, while the proposed label measures total surplus. A correct welfare ranking need not change a trader's choice. P trains agents with potential-based rewards; Q studies prompted traders. With fixed initial and zero terminal potential, P's discounted shaping terms telescope to a constant. The argument requires an appropriate potential on a trader's exit transition. The record shows no error in P's implementation.

P's unregularised Bradley–Terry fit has a separate boundary. If all K comparisons favour one of two agents, the graph is connected, but the loss for score difference d is $K \log(1 + e^{-d})$, with no finite minimum as d grows. Ada derived this and Emmy agreed. A larger quote set would need regularisation or symmetric pseudocounts; two quotes need only a direct judgement.

The price identity holds for a fixed match and continuation; a changed quote can alter both. Crossing also changes who is eligible for Q's next random draw. Replays must couple underlying draws but apply the selection rule in each branch. P notes that hidden, domain-specific state can be outside its scope. This example quantifies that limit; it neither refutes P nor establishes how often these auction states arise.

5 Why the thread stopped

The verifier accepted the fixed-schedule, hidden-value construction, its explicit continuation, and the distinction between Q's quote actions and P's agents. This justified DRAFT as a theoretical result. No auction replay measured state frequency, judge accuracy, or any gain over direct conditional-welfare prediction. A pilot could restart the thread: find Q states with both quotes legal, replay each under coupled draws and fixed continuation policies, then compare local-information judgements with direct prediction and a more informed reference.